Konrad Dierks

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Personal Information

Citizenship: German
Gender: Male

TSE Placement

Placement Officers

Professor Paul Diegert
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Professor Mathias Reynaert
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Louise Strachan
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Current Position

PhD student at Toulouse School of Economics, expected completion 2027
Advisor: Christian Gollier
Job Market Paper: "Is a 100% renewable electricity mix cost-efficient under weather uncertainty?"

References

Professor Christian Gollier
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Professor Fabrice Collard
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Professor Stefan Ambec
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Professor Ingela Alger
Toulouse School of Economics
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Research Interests

Main: Energy economics
Secondary: Political economics, optimization, and numerical methods

Prior Education

2022–2023	Toulouse School of Economics MRes in Economics
2021–2022	Toulouse School of Economics Master in Economic Theory and Econometrics, very good
2019–2021	ENS de Lyon (France) Master in Advanced Mathematics, probability theory and statistics track, good
2018	Università degli Studi di Firenze (Italy) Erasmus+ semester
2015–2019	Technische Universität Dresden (Germany) Bachelor in Mathematics, minor in Economics, very good

Research Papers

"Is a 100% renewable electricity mix cost-efficient under weather uncertainty?" (Job Market Paper)

The generation of wind and solar power is intermittent and uncertain, posing challenges to decarbonizing the electricity mix. I propose a dynamic stochastic model of wind and solar generation and optimal storage operation at the national level. The model is calibrated on German data and accounts for both flexible consumption and demand curtailment. Uncertainty and constraints on storage levels preclude an analytical solution. Evaluating the optimal operation numerically, I optimize over renewable and storage capacities to obtain the efficient electricity mix. With battery storage only and a 5% discount rate, the optimal 100% renewable mix features a welfare-adjusted levelized cost of 252 €/MWh. In spite of its large round-trip losses, allowing for hydrogen storage reduces the optimal renewable capacities sharply and lowers this cost to 119 €/MWh: batteries are prohibitively costly for providing the large stockpile required. Relative to perfect foresight over the observed decade, weather uncertainty mainly raises this stockpile, from 22 to 35 TWh of hydrogen storage, with small effects on other capacities and the levelized cost. The efficient mix is such that large amounts of electricity are produced with near-zero marginal value in many states. With a fossil backup, a non-negligible storage capacity only becomes optimal at a positive carbon price, and marginal abatement costs rise sharply close to zero emissions if there are no other low-carbon alternatives.

"Does universalization ethics justify participation in large elections?"

Joint with Ingela Alger and Jean-François Laslier; under review at *Theoretical Economics*
Available as a TSE working paper

Why would voters incur costs to cast votes they believe are inconsequential? We show that a form of universalization ethics can help explain this "paradox of costly voting." The political model has two candidates, a continuum of voters, a known underdog, and two novel elements: core voters (who always vote), and distinct election stakes for the partisan groups. Elections are modeled as power-sharing institutions, winner-take-all being a limit case. Cost-sensitive voters, whose turnout behavior we analyze, consider the election outcome, should their decision be universalized to other voters. The "others" can be limited to co-partisans ("partisan ethics") or not ("nonpartisan ethics"). In contrast to most of the literature, we do not impose conditions ensuring existence and uniqueness.
The universalization ethics often fosters participation, but coordination problems remain. Equilibrium can fail to exist under partisan ethics, and equilibrium multiplicity can arise under both partisan and nonpartisan ethics. Equilibria where the underdog wins may coexist with equilibria where the topdog wins. The former are sustained because of a boost of cost-sensitive voters' willingness to vote, stemming either from a large core constituency or from a high stake. In this case, the core constituency is a complement rather than a substitute for the cost-sensitive voters.

Teaching

2025	TA for Mathematics, first-year Licence in Economics and Business, TSE, in French
2024	TA for Probability and Statistics, third-year Licence in Economics, TSE, in French
2022, 2023	TA for Microeconomics, second- and third-year Licence in Economics, Economics and Mathematics, TSE, in English and French
2017, 2018	TA for Linear Algebra, first-year Bachelor in Mathematics, TU Dresden, in German

Professional Experience

April 2019–Aug. 2019	Student job at Fraunhofer Institute for Transportation and Infrastructure Systems (IVI) in Dresden (Germany), skills: Python programming, applied statistics, optimal control

Scholarships and Funding

2026–	ANR grant for the project AVINEQ, "Inequality Aversion: Measurements and Implications for Redistributive Policies and Long-Term Investments" (PI: Christian Gollier)
2026–	Computing resources from the MesoNET resources center (allocation m26158, ANR-21-ESRE-0051)

2023–	PhD funding under the label of the BioEco Graduate School (EUR BioEco, ANR-18-EURE-0021)
2021–2022	UniCredit Foundation Masterscholarship (Master in Economic Theory and Econometrics, TSE)

Conference Talks

2026 Doctorissimes (Paris); HYRCE (Liège); EEA-ESEM Congress (Dublin); IAEE European Conference (Munich); IAEE COP Symposium (Antalya, upcoming)

2025 Interdisciplinary PhD Workshop in Sustainable Development at Columbia University (New York City, on Zoom); QMUL Economics and Finance PhD Workshop (London); ENTER Jamboree at SSE (Stockholm); EAERE Annual Conference (Bergen); FAERE Annual Conference (Nantes)

2024 EEA-ESEM Congress (Rotterdam); Economic Science Association European Meeting (Helsinki)

Other Talks

2026 EDF (Paris, TSE-EDF cooperation)

2025 ENTER Exchange Seminar at UAB (Barcelona); University of Leipzig Lunchtime Seminar for PhD Students

Languages

German (native), English (fluent), French (fluent), Italian (advanced), Spanish (basic)